

Question 1:

The key differences between routing and forwarding is that forwarding operates at the data plane. Forwarding moves the packets as a muscle from an input port to an output port on a router.

While routing operates at the control plane. It is the brain which determines the end to end path that packets should take through the network. It also creates the routing tables that the muscle, forwarding, depends on.

Question 2:

The data plane handles the actual movement of packets through a router. It decides for the datagrams which output port it should go based on the forwarding tables. Overall the data plane focuses on fast, local actions like switching packets within the network layer.

The control plane, on the other hand, is in charge of creating algorithms so that there are paths for the packets to take. It runs routing protocols that exchange information with other routers. All in all, the data plane moves packets, while the control plane decides the routes those packets should follow.

Question 3:

Destination-based forwarding means the router looks only at the destination IP address in the packet header to decide where to send the packet. Header to decide where to send the packet. Generalized forwarding is more flexible; it allows routers to make decisions based on multiple packet header fields. So the main difference is that destination-based forwarding uses only one field, while generalized forwarding uses multiple fields.

Question 4:

Your ISP assigns your wireless router a public IP when you connect it to your modem. The router then hands out private IP addresses, like 192.168.x.x or 10.0.0.x, to your five home PCs using DHCP. These private addresses work only within your home network. Since everyone shares the same public IP to reach the Internet, the router.

Now NAT maps each private IP to the public IP for external communication and reverses the mapping for replies. So yes, wireless uses NAT because it makes it possible for multiple computers to browse the Internet using one ISP-provided public IP.

Question 5:

Under link-state routing, routers exchange link information with all other routers, enabling each device to independently compute the network's shortest paths using Dijkstra's algorithms.

Distance-vector routing operates differently by building algorithms based on information received from neighboring routers.

However, both require more CPU and memory, both use algorithms to help routers determine routes, both are part of the control plan and help forwarding in the data plane.

Question 6:

No. Each autonomous system has its own administrative autonomy for choosing its internal routing algorithm, so different autonomous systems can use different routing protocols.

Q7:

(a)

The cost of the link to X is 1.

This is because X's shortest distance is 8 and Y's distance is 7, so the link cost is $8 - 7 = 1$.

(b)

The cost of the link to Y is 7.

This comes from Y's shortest distance of 7 minus Z's distance of 0, giving a link cost of $7 - 0 = 7$.

Q9:

(a)

When the Bellman–Ford algorithm converges, the shortest-path distance vector at router

Y is $(u,v,w,x,y) = (16,7,10,3,0)$

This shows that the minimum-cost path from Y to u is 16, to v is 7, to w is 10, to x is 3, and to itself (y) is 0.

(b)

The problem that occurs when link costs increase in distance-vector routing is called the count-to-infinity problem.